

Management System : Automating Billing and Enhancing Customer Experience

Submitted to:

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Description

In today's digital economy, when subscription-based services are becoming more popular, a Subscription Plan Management System assures proper billing, increases customer retention, and gives real-time insights into performance and income. (Kanuri et al.,2017) With features such as automated billing, customer self-service portals, and comprehensive monitoring, it becomes an important platform for any organization that provides recurring services, from subscription-based services to media streaming platforms. Businesses may use this tool to automate their operations, improve the customer experience, and easily scale their subscription models.

According to Gandhi et al. (2023), Digital subscriptions provide numerous business benefits. They provide a consistent source of revenue and allow firms to expand into new markets that would otherwise be out of reach for physical locations. The subscription-based service business has grown due to technological improvements, shifting customer patterns, and a need for tailored and easy experiences.

A Subscription Plan Management System is an advanced solution that allows businesses to handle recurring revenues more efficiently by automating and optimizing the whole subscription service lifecycle. It simplifies complicated tasks, such as onboarding consumers, billing, payments, renewals, and plan modifications, allowing organizations to focus on growth and customer happiness. It allows for easy management of different subscription plans, customizable billing periods, and fast transitions between plan upgrades and downgrades.

Core features

- **Customer self-service portal**

It includes the name of the user, contact information, payment method and subscription history and communication preference. Customers can manage their own subscriptions, view invoices, and update payment information. Users can change their subscription plans directly from the portal, with prorated charges or credits.

- **Plan Customization**

The ability of the client to customize the subscription plan. It helps to provide flexibility and personalization, enabling customers to choose or modify plans that best suit their requirements. They can choose what subscription plan (basic, premium, etc) and custom billing cycles (monthly or annually).

- **Subscription life cycle**

The customer's subscription begins when they join up and ends when they upgrade, downgrade, renew, or cancel it. It includes free trials with automatic transition to paid plans, it will notify you when your trial or subscription is ending.

- **Payment**

It is the process of collecting and processing payments from customers for their subscription-based services. Effective payment management is critical for ensuring timely and secure transactions, maintaining customer satisfaction, and supporting the overall subscription lifecycle. It will notify you when the payment is successful or not.

- **Customer Retention and engagements**

It includes loyalty programs that will allow long term subscribers/user with rewards, discounts and extra benefits. It also motivates users to stay longer, reducing churn

- **Notifications and alert**

help keep customers informed and engaged throughout their subscription lifecycle. It also notify the users promotional offer like upgrading your plan at a discounted price

- **Analytics and Reporting**

allowing businesses to track performance, understand customer behavior, and make informed decisions. Reporting features can help detect problems in the subscription lifecycle, such as high cancellation rates and payment problems. Analyzing data can help firms optimize operations and improve user experience.

High Level Entities

- **User**

It includes the name of the user, contact information, payment method and subscription history and communication preference(email or contact number).

- **Subscription Plan**

A subscription plan is a predefined offering that outlines the details of the subscription, including features, pricing, duration, and terms of service. It includes categorized plans(basic, premium, etc), billing intervals (per month), discount(loyalty card).

- **Subscription**

A subscription refers to the individual instance of a user's commitment to a subscription plan. It represents the user's active engagement with the service based on a specific plan.

- **Payment**

It includes payment method, transaction history, Proration calculation (upgrading/downgrading a plan, mid-cancellation(refund on request(if have a valid reason they can get a half refund or full refund)), Notification of the transaction is successful or failed(communication preference)

Business rules

- **User**

Users must have a unique UserID, contact information, payment method and subscription history and communication preference. User can only subscribe to one subscription. A user's subscription must have a unique SubscriptionID and link to a UserID.

- **Subscription Plan**

Each subscription plan must have a unique identifier(PlanID).Plans can be categorized (Basic, Standard, Premium). Each plan must have a Plan name, description, price, and billing intervals (monthly, annually).

- **Subscriptions**

Each Subscription must have a unique identifier (SubscriptionID). A SubscriptionID must be linked to UserID. The subscriptions must include startdate ,enddate and status. Subscription status can be Active, Canceled, or Expired.

- **Payment**

Each payment must have a unique PaymentID and it must be link to SubscriptionID, and it must have payment method, amount and payment date and status. Failed payments must notify the user for review and can put it into a "Paused" state.

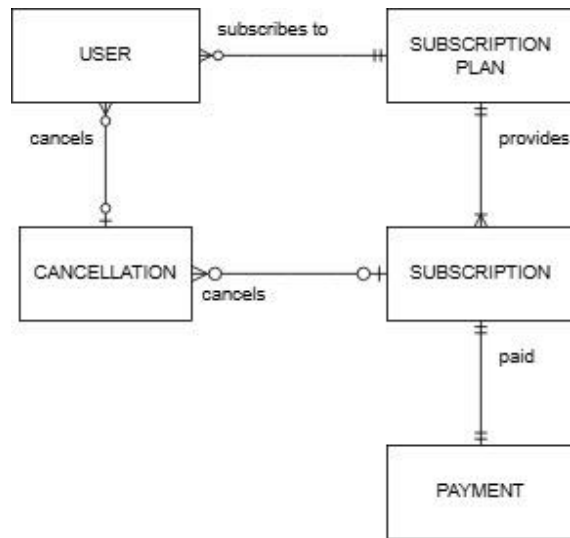
- **Discount codes**

Discount must have a unique identifier (DiscountCode), then DiscountAmount, ExpirationDate, UsageLimit Discount codes can be applied to new subscriptions.Discounts must have an expiration date and specific usage limits. A user may only apply one discount code per subscription.

- **Cancellation**

Users may cancel their subscription at any time, but they will retain access until the end of the current monthly cycle. Refunds for partial months are not permitted unless specifically stated in the terms. It must have a unique identifier (CancellationID) and it must be linked in SubscriptionID and UserID to know who is canceling the description then CancellationDate, reason and status(pending, processed, complete) and refundAmount.

Entity-Relationship Diagram (ERD)



Detailed Entity-Relationship Diagram (ERD)

A detailed Entity-Relationship Diagram (ERD) for Subscription Plan Management System has a list of attributes or properties that can be associated with a certain entity.

These are the list of attributes or properties for Subscription Plan Management System:

- **User**
 - User_ID (PK)
 - Name
 - Contact information
 - payment method
 - subscription history
 - communication preference
- **Subscription Plan**
 - Plan_ID (PK)
 - Plan name
 - Description
 - Price
 - billing intervals
- **Subscription**
 - Subscription_ID (PK)
 - User_ID (FK)
 - Startdate
 - Enddate

- Status
- **Payment**
 - Payment_ID (PK)
 - Subscription_ID(FK)
 - payment method
 - amount
 - payment date
 - Status
- **Cancellation**
 - Cancellation_ID (PK)
 - Subscription_ID(FK)
 - User_ID (FK)
 - CancellationDate
 - reason
 - Status
 - refundAmount

